# AI-Based Climate Risk Mapping for Vulnerable Communities in Northeast Syria

## Heat and Agricultural Drought Risk Assessment in Raqqa, Deir ez-Zor, and Al-Hasakah

FTL Syria AI4Climate – Group 11 | Field: Computer and Automation Engineering

## 1. Project Idea

Northeast Syria is increasingly affected by extreme heat and agricultural drought, threatening agricultural production, water availability, and the livelihoods of vulnerable communities. However, accessible, localized information to identify and compare areas exposed to these climate risks remains limited.

This project aims to develop an AI-based climate risk mapping prototype for Raqqa, Deir ez-Zor, and Al-Hasakah. Using historical climate and satellite data, it will assess and classify areas according to heat and agricultural drought risk and present results through an interactive map. The prototype will help reveal spatial risk patterns and inform climate-related planning and decision-making.

## 2. Relevance to Sustainable Development Goals (SDGs)

**SDG 13 – Climate Action:** Supports climate risk assessment and evidence-based adaptation planning.
**SDG 2 – Zero Hunger:** Identifies areas where agricultural drought may threaten food production and livelihoods.
**SDG 11 – Sustainable Cities and Communities:** Provides spatial climate-risk information to support resilient communities and local planning.

## 3. Literature Examples

1. **Sukkar et al. (2024).** *Investigating the impacts of climate variations and armed conflict on drought and vegetation cover in Northeast Syria (2000–2023).* Journal of Arid Environments, 225, 105278. This study examines drought and vegetation-cover changes in Northeast Syria, providing regional evidence relevant to climate-risk assessment.
https://doi.org/10.1016/j.jaridenv.2024.105278

2. **Sukkar et al. (2025).** *Multisource Remote Sensing and Machine Learning for Spatio-Temporal Drought Assessment in Northeast Syria.* Sustainability, 17(24), 10933. This research demonstrates the use of remote-sensing data and machine learning to assess drought patterns in the project's target region.
https://doi.org/10.3390/su172410933

**Expected Outcome:** A preliminary AI-based climate-risk classification and interactive map prototype covering Raqqa, Deir ez-Zor, and Al-Hasakah.